

Monte Carlo Method

Exercise:

1. Determine Area of irregular figure

Hence $\frac{F}{A} = \frac{M}{N}$

F = Area of irregular figure
A = Area of square to be considered

Random number between (0-1) M = Points within shape

For all cases we will consider

N = Total random points

$9 \times 9 \text{ cm}^2$ regular shape

From Random number table

$$x_1 = 0.96 \times 9 = 8.64$$

$$y_1 = 0.42 \times 9 = 3.78$$

$$x_2 = 0.71 \times 9 = 6.39$$

$$y_2 = 0.10 \times 9 = 0.9$$

$$x_3 = 0.03 \times 9 = 0.27$$

$$y_3 = 0.30 \times 9 = 2.7$$

According to basic mathematical equation

$$\text{Area} = \frac{26}{84} \times (9 \times 9) = 25.07 \text{ cm}^2$$

$$\therefore \text{Area of square} = 9 \times 9 = 81 \text{ cm}^2$$

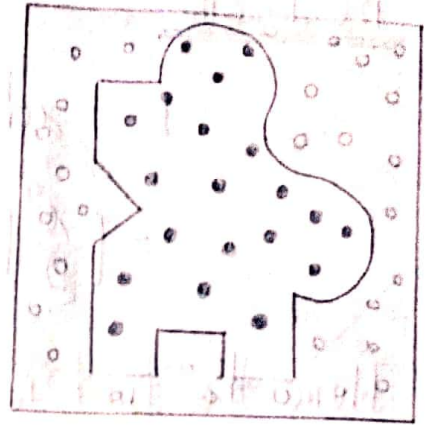
Considering the ratio of points falling into range which is 30.95%

Case 1:

$$\text{Area, } A = \frac{20}{50} \times 81 = 32.4 \text{ cm}^2$$

$$\text{Main area} = 25.07 \text{ cm}^2$$

$$\therefore \text{Difference} = (32.4 - 25.07) \\ = 7.33$$

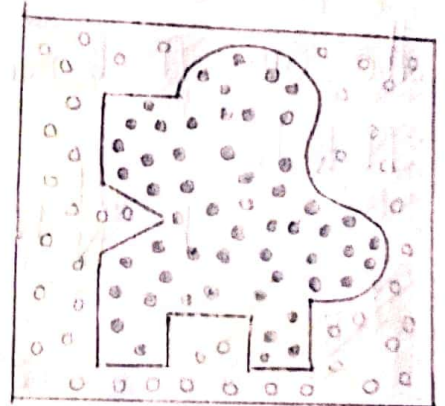


Case 2:

$$\text{Area, } A = \frac{36}{100} \times 81 = 29.16 \text{ cm}^2$$

$$\text{Calculated area} = 25.07 \text{ cm}^2$$

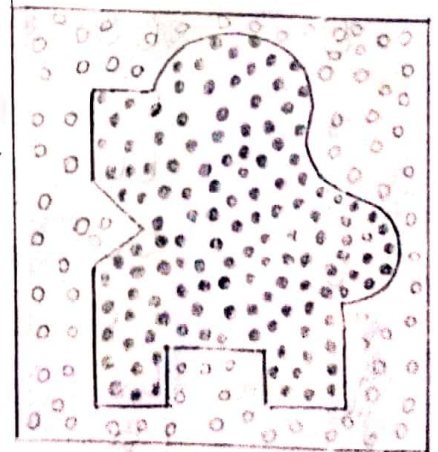
$$\therefore \text{Difference} = (29.16 - 25.07) \text{ cm}^2 \\ = 4.09 \text{ cm}^2$$



Case 3:

$$\text{Area, } A = \frac{65}{200} \times 81 = 26.325 \text{ cm}^2$$

$$\text{Difference} = (26.325 - 25.07) \text{ cm}^2 \\ = 1.255 \text{ cm}^2$$



Accuracy grows higher with increasing points

2. Integrals by Monte Carlo

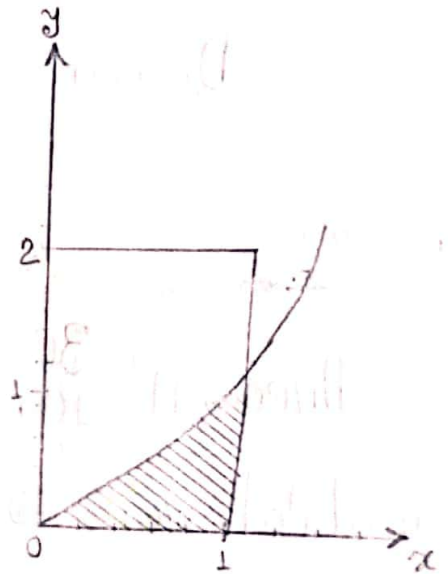
i) Solution:

$$I = \int_0^1 x dx$$
$$= \left[\frac{x^2}{2} \right]_0^1 = \frac{1}{2} \quad (\text{The exact value of interval})$$

Here the limit is between $(0-1)$

The ratio of the shaded area I to the area A of rectangle $ABCD$ is M to N

$$I = \frac{M}{N} A$$



Simulation of $I = \int_0^1 x dx$

Random Number	x $0.1(\pi)$	Random Number	y $I(\pi)$	M	N
9	0.9	0.57	0.57	0	1
7	0.7	0.18	0.18	1	2
6	0.6	0.90	0.90	1	3
6	0.6	0.76	0.76	1	4
5	0.5	0.42	0.42	3	5
3	0.3	0.77	0.77	3	6
5	0.5	0.18	0.18	4	7
2	0.2	0.8	0.8	4	8
4	0.4	0.0	0.0	5	9

x has random raising in between 0-10 and y to 0-1.

$$I = \frac{5}{9} (2 \times 1) = 1.5 \text{ which is quite the exact value}$$

ii) Solutions

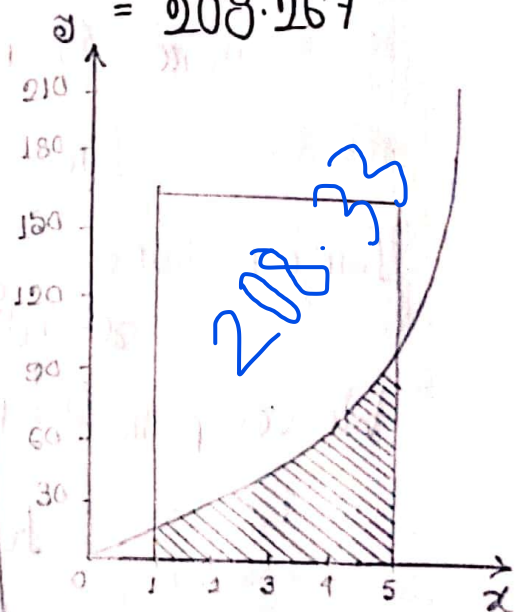
Given that

$$I = \int_1^5 \frac{x^4}{3} dx = \int_a^b f(x) dx$$

The exact value of integral, $I = \left[\frac{x^5}{15} \right]_1^5 = \left(\frac{3125 - 1}{15} \right) = 208.267$

Simulation of $I = \int_1^5 \frac{x^4}{3} dx$

Random Number	x 0:1(π)	Random Number	y 0:1(π)	$\frac{x^4}{3}$	M	N
12	1.2	.57	99.75	.69	0	1
19	1.9	.18	31.05	4.3	0	2
23	2.3	.00	00	9.3	1	3
35	3.5	.27	47.25	50.2	2	4
41	4.1	.42	73.5	94.2	3	5
71	1.1	.45	78.75	0.49	3	6
13	1.3	.87	152.3	0.95	3	7
20	2.0	.25	43.8	5.33	3	8
29	2.9	.10	17.5	23.57	3	9
37	3.7	.60	105	62.5	3	10
47	4.7	0.77	115.5	162.7	4	11
33	3.3	0.26	35.75	39.5	5	12
25	2.5	0.05	8.75	13.02	6	13
26	2.6	0.40	70	15.23	6	14



We can't draw the rectangle which is less than the $f(x)$ maximum value.

Here the x has random generated 0-50 and for y its opted to be 0-1. For $x = 0.1$ (Random)
 $y = 175$ (Random)

$$I = \left(\frac{4}{14}\right) (175 \times 4)$$

$= 199.99 \approx 208.267$ which is near to the approximate value.

3. Determine the value of P_i

Here is given that

$$\text{Area of circle} = \frac{\pi D^2}{4} = \frac{\pi (2r)^2}{4} = \pi r^2$$

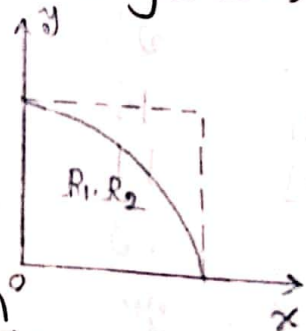
For equation

$$x^2 + y^2 \leq 1 \quad \because x, y \geq 0$$

We've pair of Random number R_1, R_2 in range (0,1)

$$R_1^2 + R_2^2 \leq 1$$

$$\therefore R_2 \leq \sqrt{1 - R_1^2}$$



M/M will take approach and get value $\frac{\pi}{4}$

Area under the curve

R_1	R_2	$\sqrt{1-R_1^2}$	I/O	R_1	R_2	$\sqrt{1-R_1^2}$	I/O
.82	.95	.5724	0	.76	.18	.6499	1
.34	.14	.9404	1	.33	.81	.9440	1
.20	.84	.9797	1	.35	.96	.9368	0
.58	.14	.8146	1	.29	.57	.9570	1
.52	.03	.8542	1	.95	.41	.3123	0
.51	.60	.8602	1	.84	.51	.5426	1
.34	.38	.9404	1	.34	.68	.9404	1
.72	.11	.6940	1	.85	.11	.5268	1
.50	.55	.8760	1	.58	.92	.8146	0
.08	.21	.9967	1	.69	.59	.7238	1

$$\text{For 20 points} = \frac{16}{20} \times 4 = 3.2$$

R_1	R_2	$\sqrt{1-R_1^2}$	I/O	R_1	R_2	$\sqrt{1-R_1^2}$	I/O
.82	.95	.5724	0	.85	.11	.5268	1
.34	.14	.9404	1	.58	.92	.8146	0
.20	.84	.9797	1	.69	.59	.7238	1
.58	.14	.8146	1	.48	.37	.8773	1
.52	.03	.8542	1	.51	.72	.8602	1
.51	.60	.8602	1	.06	.33	.9982	1
.34	.38	.9404	1	.22	.90	.9755	0
.72	.11	.6940	1	.80	.76	.6000	1
.50	.55	.8760	1	.56	.29	.8265	1
.08	.21	.9967	1	.06	.66	.9982	0
.76	.18	.6499	1	.92	.40	.3919	1
.33	.81	.9440	1	.51	.11	.8602	1
.35	.96	.9368	0	.13	.44	.9915	1
.29	.57	.9570	1	.65	.74	.9599	1
.95	.47	.3123	0	.60	.27	.8000	1
.84	.51	.5426	1	.51	.16	.8602	1
.34	.68	.9404	1	.50	.41	.8602	1
				.13	.20	.9915	1

R_1	R_2	$\sqrt{1-R_1^2}$	1/0	R_1	R_2	$\sqrt{1-R_1^2}$	1/0
.94	.68	.3412	0	.51	.95	.8216	0
.26	.28	.9656	1	.78	.75	.6258	0
.33	.16	.9440	1				

For 40 points, $\pi = \frac{31}{40} \times 4 = 3.10$

4. Pi value determining program:

```

import math
import random
import matplotlib.pyplot as plt

random.seed(1)

N=1000
hit=0
interval=1/1000
ax=[] ay=[]

for i in range(1,1001):
    x=i*interval
    ax.append(x)
    y=math.sqrt(1-x*x)
    ay.append(y)

plt.figure(figsize=(6,6))
plt.plot([0,1,1,0],[0,0,1,1],c='blue')

plt.plot(ax,ay,c='blue')

hx=[] hy=[]
mx=[] my=[]

```

```

for i in range (1, N+1):
    x = random.uniform (0, 1)
    y = random.uniform (0, 1)
    if y <= math.sqrt (1-x*x)
        hit += 1
        hx.append (x)
        hy.append (y)
    else:
        mx.append (x)
        my.append (y)

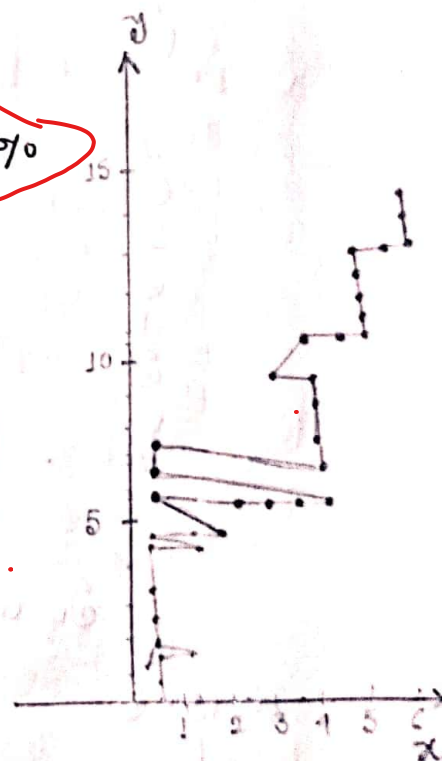
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$est_pi = hit * 4 / N$
 print (est_pi)

5. Random walk Problem

The problem associated with these 40%, 10%, 25%, 25% forward, backward, left, right respectively covering 75, 45, 60, 60 cm

Direction	Probability	Random Num
Forward	0.4	0, 1, 2, 3
Backward	0.1	4
Right	0.25	5, 6, 7
Left	0.25	8, 9



Random Num	Direction	Position		Random	Direction	Position	
		x	y			x	y
6	R	.6	0	5	R	3.6	6.3
2	F	.6	.75	7	R	3.2	6.3
0	F	.6	1.5	4	B	4.2	5.55
6	R	1.2	1.5	2	F	4.2	6.3
8	L	.6	1.5	0	F	4.2	7.05
4	B	.6	1.05	1	F	4.2	7.8
5	R	1.2	1.05	2	F	4.2	8.55
9	L	.6	1.05	3	F	4.2	9.3
3	F	.6	1.8	9	L	3.6	9.3
2	F	.6	2.55	5	R	4.2	9.3
2	F	.6	3.3	1	F	4.2	10.05
1	F	.6	4.05	3	F	4.2	10.8
7	R	1.2	4.05	4	B	4.2	10.05
8	L	.6	4.05	6	R	4.8	10.05
7	R	1.2	4.05	7	R	5.4	10.05
6	R	1.8	4.05	1	F	5.4	10.8
1	F	1.8	4.8	2	F	5.4	11.55
9	L	1.2	4.8	3	F	5.4	12.3
0	F	1.2	5.55	5	R	6.0	12.3
2	F	1.2	6.3	6	R	6.6	12.3
3	F	1.2	7.05	9	L	6.0	12.3
4	B	1.2	6.3	1	F	6.0	13.05
5	R	1.8	6.3	1	F	6.0	13.8
6	R	2.4	6.3	4	B	6.0	13.05
7	R	3.0	6.3	5	R	6.6	13.05

So distance travelled 13.05 in y axis & 6.6 in x axis.

G Bombing Problems Simulation:

$$x_1 = 5002x$$

$$y_1 = 3002y$$

RNN	x	RNN	y	Re	RNN	x	RNN	y	Result
.23	115	.24	72	H	.13	65	.15	45	H
-1.16	-580	-0.02	-6	M	.02	10	-.43	-129	H
0.39	195	0.64	192	H	-1.05	-525	-.57	-171	M
-1.90	-950	-1.04	-312	M	-.55	-275	-.52	-156	M
-0.78	-390	0.68	204	H	.9	450	-.13	-39	H
-0.02	-10	-0.47	-141	H	1.16	580	1.13	339	H
-0.40	-200	-0.75	-225	H	.24	120	1.02	306	M
-0.66	-330	-0.44	-132	H	.37	185	-0.2	-60	H
1.41	705	1.21	363	M	-.95	-475	.93	279	M
0.07	35	-0.08	-24	H	-.03	-15	-1.01	-303	H
0.53	265	-1.56	-468	M	-1.17	-585	-.15	-45	M
0.03	15	1.49	447	M	-.68	-340	-.21	-63	M
-1.19	-595	-1.19	-357	M	.25	125	1.53	459	M
0.11	55	-1.86	-558	M	.17	85	.14	42	H
0.16	80	1.21	363	M	1.11	555	.16	48	H
-0.82	-410	0.75	225	H	.20	100	1.17	351	M
0.42	210	-1.50	-450	M	-.35	-175	-1.18	-354	H
-0.89	-445	0.19	57	H	.5	250	.29	87	H
0.49	249	-1.44	-432	M	-1.00	-500	.30	90	M
-0.21	-105	0.65	195	H	-.53	-265	.52	156	M

Number of hit = 20

Number of miss = 20

 $\therefore$ % of strike = 50

7. Random Variable:

$$x \sim N(\mu=50, \sigma^2=65)$$

$$y \sim N(\mu=100, \sigma^2=100)$$

$$z \sim N(\mu=150, \sigma^2=150)$$

RN1	x	RN2	y	RN3	z	x+y	w
✓0.46	20.1	✓0.09	109	✓0.59	238.5	129.1	0.54
-1.05	-18.3	0.56	156	-0.67	49.5	137.7	2.78
-0.15	40.3	-0.02	98	0.41	211.5	138	0.65
-0.81	-2.65	3.10	910	0.51	226.5	407.35	1.79
-0.74	1.9	-0.44	56	1.53	379.5	57.9	0.15
-0.39	24.65	-0.33	67	-0.37	94.5	91.65	0.96
-0.45	20.75	0.40	140	-0.27	109.5	160.75	1.46
-2.44	-108.6	0.38	138	-0.15	127.5	29.4	0.23

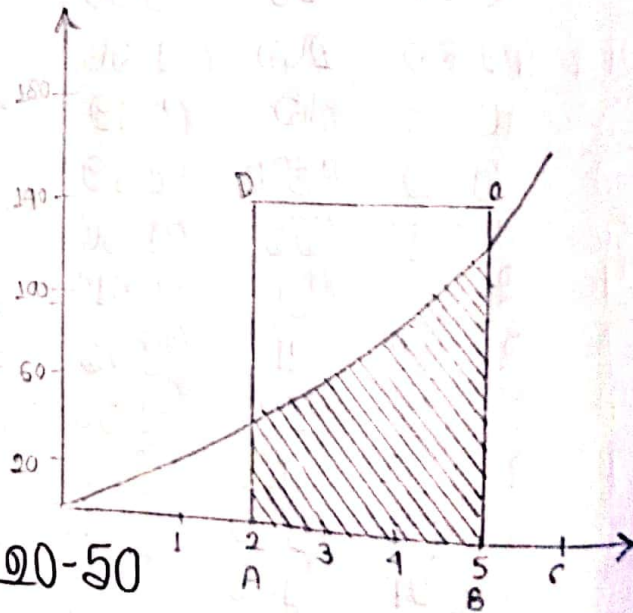
8. Integral By Monte Carlo

$$I = \int_a^b x^3 dx = \int_a^b f(x) dx$$

$$= \left[\frac{x^4}{4} \right]_2^5 = 152.25$$

Ratio of ABCD shaded area $I = \frac{N}{N} A$

Here x Ranges random number 20-50
and y (0-1) if $y \leq x^3$ then count the
number in range N



Random Number	x 0.1π	Random Number	y $140(\pi)$	x^2	M	N
22	2.2	.57	79.8	10.65	0	1
25	2.5	.18	25.2	15.63	0	2
18	1.8	.00	0.0	5.83	1	3
45	4.5	.90	126.0	91.13	1	4
25	2.5	.05	7.00	15.63	2	5
27	2.7	.77	107.8	19.68	2	6
48	4.8	.66	92.4	110.60	3	7
43	4.3	.10	14.0	79.51	4	8
40	4.0	.76	106.4	64.0	4	9
47	4.7	.42	58.8	103.83	5	10
38	3.8	.78	109.2	54.87	5	11
33	3.3	.88	123.2	35.94	5	12
24	2.4	.03	4.2	13.82	6	13
47	4.7	.09	12.6	103.80	7	14
42	4.2	.77	107.6	74.09	7	15
25	2.5	.61	85.4	15.63	7	16

$$I = \left(\frac{7}{16}\right) \times (140 \times 3) = 183.75$$

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10. Bearing Problem:

Hours	Probability	Cum-Prob	Random digit
1200	.10	.10	01-10
1400	.25	.35	11-35
1600	.30	.65	36-65
1800	.25	.90	66-90
2000	.10	1.0	91-00

Delay time	Probability	Cum-Prob	Random Digit
4	0.3	0.3	1-3
6	0.6	0.9	4-9
8	0.1	1.0	0

RD	Life	Clock	RD	Delay	RD	Life	Clock	RD	Delay	RD	Life	Clock	RD	Delay
25	1400	1400	9	6	65	1600	1600	5	6	94	2000	2000	7	6
82	1800	3200	8	6	91	2000	3600	0	8	87	1800	3800	0	6
16	1400	4600	0	8	30	1400	5000	5	6	63	1600	5400	6	6
14	1400	6000	5	6	66	1800	6800	6	6	66	1800	7200	2	4
82	1800	7800	2	4	32	1600	8400	5	6	30	1400	8600	2	4
45	1600	9400	3	4	29	1400	9800	3	4	69	1800	10400	7	6
81	1800	11200	7	6	11	1400	11200	2	4	37	1600	12000	2	4
80	1800	13000	3	4	43	1600	12800	8	6	1	1200	13200	8	6
97	2000	15000	7	6	40	1600	14400	0	8	66	1800	15000	8	8
62	1600	16600	4	6	65	1600	16000	8	6	51	1600	16600	3	6
30	1400	18000	9	6	82	1800	17800	2	4	92	2000	18600	8	6
31	1600	19600	5	6	73	1800	19600	4	6	36	1600	20200	3	4
80	1800	21400	6	6	15	1400	21000	4	6	47	1600	21800	7	6
72	1800	23200	7	6	70	1800	22800	6	4	80	1800	23600	0	4
Total Delay			1260					12880					12976	

$$\text{Delay from calculation} = 80 + 80 + 76 \\ = 236$$

$$\text{Cost of bearing} = 69 \times 14 = 966$$

$$\text{Total downtime} = 966 + 236 = 1202$$

$$\text{Cost of downtime} = 1202 \times 5 = 6010$$

$$\text{Cost of repairment} = 1380 \times 1202 \times \frac{19}{42} = 543.76$$

11. Reliability Problem Program

```
import math
import random
import matplotlib.pyplot as plt

n = 100000
Delay = 0
Life = 0

for i in range(1, n):
    RD1 = random.randint(1, 100)
    RD11 = random.randint(0, 9)

    if 1 < RD1 < 10:
        Life = Life + 1000
    elif 11 < RD1 < 24:
        Life = Life + 1100
    elif 25 < RD1 < 48:
        Life = Life + 1200
```

```
    elif 49 < RD1 < 62:
        Life = Life + 1300
    elif 63 < RD1 < 74:
        Life = Life + 1400
    elif 75 < RD1 < 84:
        Life = Life + 1500
    elif 85 < RD1 < 90:
        Life = Life + 1600
    elif 91 < RD1 < 95:
        Life = Life + 1700
    elif 96 < RD1 < 98:
        Life = Life + 1800
    else:
        Life = Life + 1900

    if 1 < RD11 < 3:
        delay = delay + 4;
```


elif $4 < RD11 < 9$: delay = delay + 6 else: delay = delay + 8	Printf (delay) Printf (Life)
--	---------------------------------

13. Hinge Assembly Problem

```

import math
import random
import matplotlib.pyplot as plt

n = 1000, count = 0
for i in range(1, n):
    RDa = random.rand(0,1)
    RDb = random.rand(0,1)
    RDc = random.rand(0,1)
    RDd = random.rand(0,1)

    R = RDa(0,1) + 1.95
    b = RDb(0,1) + 1.95
    c = RDc + 29.5    d = 34 + RDd
    Sum = a + b + c

    if sum > d:
        count++

Result = count / n * 100
Print (Result)

```